

RAS AL KHAIMAH INTL, United Arab Emirates

WMO: 411840

Lat: 25.613N

Lon: 55.939E

Elev: 102

StdP: 14.64

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 50.5	(c) 53.3	(d) 35.5	(e) 30.5	(f) 79.5	(g) 39.2	(h) 35.4	(i) 77.7	(j) 19.0	(k) 73.4	(l) 16.7	(m) 73.7	(n) 2.4	(o) 210	(p) 0.400

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.9	112.5	76.0	110.4	76.8	108.7	77.3	87.2	100.2	86.0	98.9	84.8	98.2	10.6	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
84.4	181.2	94.4	82.8	171.6	94.2	81.8	166.1	94.0	52.5	100.7	50.8	99.1	49.5	98.0	97.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 18.3	(b) 15.8	(c) 13.9	(e) 45.7	(f) 116.5	(g) 2.5	(h) 1.6	(i) 43.9	(j) 117.6	(k) 42.4	(l) 118.6	(m) 41.0	(n) 119.4	(o) 39.2	(p) 120.6		
(4) 18.3	(5) 15.8	(6) 13.9	(7) 45.7	(8) 116.5	(9) 2.5	(10) 1.6	(11) 43.9	(12) 117.6	(13) 42.4	(14) 118.6	(15) 41.0	(16) 119.4	(17) 39.2	(18) 120.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	83.0	66.0	68.8	74.3	82.3	90.5	94.7	97.8	97.2	92.5	85.5	76.7	69.1		
	DBStd	11.79	3.60	4.66	4.83	4.82	3.69	2.80	2.47	2.38	2.72	3.16	3.93	3.84		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	51	30	13	0	0	0	0	0	0	0	0	0	8		
	CDD50	12055	496	526	754	970	1254	1340	1483	1462	1274	1101	802	593		
	CDD65	6631	61	119	289	520	789	890	1018	997	824	636	352	136		
	CDH74	98304	483	961	2804	6531	12089	14767	17504	17003	12931	8489	3620	1122		
	CDH80	65040	53	241	1179	3793	8071	10505	13044	12544	8742	5063	1580	225		
(14) Wind	WSAvg	5.5	4.5	5.4	5.7	5.6	6.1	6.2	6.7	6.5	5.4	4.6	4.4	4.3		
(15) Precipitation	PrecAvg	4.5	0.9	1.0	1.2	0.2	0.0	0.0	0.1	0.0	0.0	0.1	0.2	0.7		
	PrecMax	11.9	4.4	5.7	6.3	1.1	0.1	0.1	1.1	0.2	0.2	1.6	2.4	6.0		
	PrecMin	1.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	3.1	1.1	1.2	1.5	0.3	0.0	0.0	0.2	0.0	0.0	0.3	0.5	1.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.1	91.0	98.2	105.8	111.4	114.6	114.8	113.5	110.0	104.0	95.3	87.9		
		MCWB	65.2	66.1	67.9	70.8	74.0	74.5	75.0	75.9	75.2	75.2	70.3	68.2		
	2%	DB	81.2	85.9	93.5	102.2	108.9	111.4	111.9	111.4	107.5	101.6	93.1	84.5		
		MCWB	65.0	66.2	66.9	70.0	73.3	75.4	76.9	77.4	76.0	74.6	70.8	67.5		
	5%	DB	78.9	82.6	89.9	99.5	106.2	109.2	109.7	109.4	105.5	99.0	91.0	82.5		
		MCWB	64.7	65.7	66.9	69.4	73.3	76.6	78.3	78.2	76.6	74.7	71.5	67.3		
	10%	DB	76.8	79.9	86.4	96.5	104.0	107.2	107.8	107.4	103.6	97.0	88.1	80.4		
		MCWB	64.1	65.2	66.7	69.4	73.8	77.2	79.2	79.1	77.7	74.8	71.2	66.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.9	72.1	74.2	76.8	83.6	87.1	89.9	88.7	87.3	84.4	77.6	72.9		
		MCDB	78.2	82.6	86.7	92.1	99.9	101.8	102.7	101.1	97.8	94.2	87.8	80.9		
	2%	WB	68.3	69.9	72.2	74.9	80.8	85.2	87.6	86.8	85.4	82.4	76.0	71.0		
		MCDB	76.7	79.1	83.9	90.8	97.6	99.3	101.2	99.2	96.6	91.9	86.3	80.0		
	5%	WB	66.8	68.4	70.7	73.5	78.9	83.9	86.2	85.4	84.2	80.8	74.6	69.7		
		MCDB	75.4	77.9	81.5	90.0	96.3	98.4	99.9	98.6	95.4	90.9	85.3	78.9		
	10%	WB	65.4	67.0	69.3	72.2	77.3	82.4	84.9	84.2	82.9	79.4	73.0	68.2		
		MCDB	74.0	76.4	79.9	88.2	95.3	97.2	98.9	98.6	94.4	90.0	84.3	77.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	21.4	21.8	22.9	26.3	27.9	26.1	21.9	21.8	24.1	25.7	23.6	22.2		
		MCDBR	24.5	25.3	28.7	31.1	30.8	29.5	25.0	25.4	27.0	27.1	25.8	25.0		
		MCWBR	12.3	11.6	11.6	11.4	12.0	12.2	10.9	11.1	11.4	11.8	12.0	12.8		
	5% WB	MCDBR	21.0	20.9	22.1	25.9	27.5	24.7	20.1	20.5	23.1	24.6	24.0	22.2		
		MCWBR	11.6	11.1	10.3	11.3	12.7	12.0	10.2	10.5	10.5	11.2	12.1	12.1		
(40) Clear-Sky Solar Irradiance	taub		0.449	0.514	0.575	0.609	0.607	0.664	0.750	0.708	0.619	0.533	0.467	0.448		
	taud		2.091	1.912	1.770	1.720	1.725	1.653	1.549	1.620	1.748	1.904	2.070	2.094		
	Ebn at Noon		249	241	234	230	230	216	198	206	220	233	242	243		
	Edh at Noon		45	58	70	76	75	80	89	83	71	58	46	43		
(44) All-Sky Solar Radiation	RadAvg		1257	1504	1796	2082	2335	2371	2211	2092	1938	1694	1365	1190		
	RadStd		88	121	127	128	52	51	56	56	30	30	42	80		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	-3.53	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (15 neighbors)	+0.74	+1.08	N/A	N/A	N/A	N/A	N/A	N/A	+234	+225

Nomenclature: See separate page